CSE-400 Scribe - L19

L19.pdf
PDF

Generate a lecture scribe intended strictly as exam-oriented reference material for CSE 400. Use only the provided context (lecture slides/pdfs and the explicitly relevant textbook parts). Treat this context as authoritative and complete. Do not use or introduce outside knowledge.

When processing handwritten notes, read them carefully and conserve their meanings. Interpret symbols, notation, and equations with precision. Resolve ambiguities strictly using consistency with the lecture slides and textbook content. Draw figures/diagrams when needed. Do not guess or hallucinate unclear material.

Write in a formal, academic tone suitable for closed-notes exam revision. Preserve the lecture's structure, ordering, notation, terminology, and scope exactly. While writing the scribe, reason step by step through the material exactly as presented in the context, and include that reasoning explicitly in the scribe.

Emphasize clarity of dependencies between assumptions, definitions, and results without introducing new explanations.

Focus more on: 1-State and explain all definitions, notation, and assumptions explicitly, 2-Present theorems, propositions, and results exactly as given, 3-Present proofs and derivations step by step, preserving all logical dependencies and, 4-Present worked examples step by step, showing all intermediate steps exactly as taught.

Content constraints: 1-Include only material present in the provided context, 2-Do not introduce new examples, interpretations, intuitions, alternative arguments, or additional explanations, 3-Do not omit conditions, assumptions, or logical steps present in the source material and, 4-Do not simplify, generalize, or paraphrase beyond faithful reconstruction.

Organization Note: Structure the scribe with clear sections and headings reflecting the lecture flow so that a student relying solely on this document can accurately reconstruct definitions, derivations, proofs, and example solutions for exam preparation.

CSE 400 — Lecture 19

N-Variate Joint PDF / PMF / CDF

(Based strictly on Lecture 19 slides)

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1. Joint Distribution of N Random Variables

Definition (Vector Representation)

For random variables $X_1, X_2, \ldots, X_N$, define the vector:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- The **joint distribution** describes the probability behavior of all variables together.
- It can be represented using:
  - Joint PMF (discrete case)
  - Joint PDF (continuous case)
  - Joint CDF

(From slide page 17)

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2. Joint Gaussian Random Variables

Joint Gaussian PDF (Two Variables)

For two random variables X and Y, the joint PDF is:

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\right.\right.\right.$$

Parameters

- μ_X, μ_Y: means
- σ_X, σ_Y: standard deviations
- ρ: Pearson correlation coefficient

(From slide pages 3 and 5)

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3. Example 1: Joint Gaussian Variables

Given

$$\mu_X = 2, \quad \sigma_X^2 = 1$$

$$\mu_Y = -3, \quad \sigma_Y^2 = 4$$

$$\mathrm{cov}(X,Y) = -1$$

Required

(a) Joint PDF $f_{XY}(x, y)$
(b) Marginal PDFs $f_X(x), f_Y(y)$
(c) $\Pr(X < 0)$ in terms of Q-function

(From slide page 4)

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4. Solution (a): Joint PDF Construction

Step 1: Correlation Coefficient

$$\rho_{XY} = \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}$$

From given:

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Substitute:

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

(From slide pages 7–8)

Step 2: Substitute into General Formula

Use:

$$\mu_X = 2, \quad \sigma_X = 1, \quad \mu_Y = -3, \quad \sigma_Y = 2, \quad \rho = -0.5$$

Substitute into joint Gaussian PDF formula.

Final Joint PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

(From slide page 10)

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5. Solution (b): Marginal PDFs

Definition

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx$$

(From slide page 11)

Key Property

If X and Y are jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

Final Marginals

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-2)^2}{2}\right)$$
$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(y+3)^2}{8}\right)$$

(From slide page 12)

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6. Solution (c): $\Pr(X < 0)$

Step 1: Identify Distribution

$$X \sim \mathcal{N}(2, 1)$$

(From slide page 13)

Step 2: Express as CDF

$$\Pr(X < 0) = F_X(0)$$

Step 3: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$
$$\Pr(X < 0) = \Pr\left(Z < \frac{0-2}{1}\right) = \Pr(Z < -2)$$

(From slide page 14)

Step 4: Symmetry Property

$$\Phi(-x) = 1 - \Phi(x) = \Pr(Z > x)$$

$$\Pr(Z < -2) = \Pr(Z > 2)$$

(From slide page 15)

Step 5: Q-function Relation

$$Q(x) = \Pr(Z > x)$$

Final Answer

$$\Pr(X < 0) = Q(2)$$

(From slide page 16)

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7. Application: Large MIMO System

System Setup

- M: transmit antennas
- N: receive antennas

(From slide pages 18–19)

Channel Coefficients

Each path between transmit antenna j and receive antenna i:

$$h_{ij} \text{ is a random variable}$$

(From slide page 20)

Channel Matrix

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1N} \\ h_{21} & h_{22} & \cdots & h_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ h_{M1} & h_{M2} & \cdots & h_{MN} \end{bmatrix}$$

- Dimension: $M \times N$

(From slide page 21)

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8. Joint PMF (Discrete Case)

Definition

For discrete random variables $X_1, X_2, \ldots, X_N$:

$$P_{X_1,X_2,\ldots,X_N}(x_1, x_2, \ldots, x_N) = \Pr(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)$$

(From slide page 22)

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9. Summary of Dependencies

- Joint distribution → defines full multivariate behavior
- Covariance → used to compute correlation ρ
- Correlation → required in joint Gaussian PDF
- Joint PDF → integrated to obtain marginals
- Marginals → used for probability computations
- Standardization → converts to standard normal
- Symmetry → enables Q-function representation